GNN-SKAN: Advancing Molecular Representation Learning with SwallowKAN

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Abstract

Effective molecular representation learning is crucial for advancing molecular property prediction and drug design. Mainstream molecular representation learning methods rely on Graph Neural Networks (GNNs) to model complex molecular structures. However, these GNN-based approaches struggle with three significant issues: insufficient annotations, molecular diversity, and architectural limitations such as over-squashing, which leads to the loss of critical structural details. To address these issues, we introduce a new class of GNNs that integrates the Kolmogorov-Arnold Networks (KANs), known for their robust data-fitting capabilities and high accuracy in small-scale AI + Science tasks. By incorporating KANs into GNNs, our model enhances the representation of molecular structures. We further advance this approach with a variant called **SwallowKAN** (**SKAN**), which employs adaptive Radial Basis Functions (RBFs) as the core of the non-linear neurons. This innovation improves both computational efficiency and adaptability to diverse molecular structures. Building on the strengths of SKAN, we propose a new class of GNNs, **GNN-SKAN**, and its augmented variant, **GNN-SKAN+**, which incorporates a SKAN-based classifier to further boost the performance. Experiments across 6 classification datasets, 6 regression datasets, and 4 few-shot learning datasets demonstrate that our approach achieves new state-of-the-art performance in terms of accuracy and computational cost.

CCS Concepts

• **Applied computing** → **Chemistry**; • **Computing methodologies** → **Reasoning about belief and knowledge**.

Keywords

Graph-based Machine Learning, Molecular Representation Learning, Kolmogorov-Arnold Networks

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1 Introduction

Precisely predicting the physicochemical properties of molecules [29, 32], drug activities [11, 38], and material characteristics [22, 35] is essential for advancing drug discovery and material science. The success of these predictions relies heavily on the quality of molecular representation learning, which transforms molecular structures into high-dimensional representations [14, 56]. However, there are two major challenges in this field: 1) insufficient labeled data due to low screening success rates [51]; and 2) substantial variations in molecular structures, compositions, and properties, which reflect molecular diversity [19].

In molecular representation learning, mainstream GNN-based methods [8, 14, 34, 57, 59] model the topological structures of molecules by representing atoms as nodes and bonds as edges. Most approaches commonly utilize the message-passing (MP) mechanism [18], commonly known as MP-GNNs, to aggregate and update node representations across molecular graphs. Notable efforts include Graph Convolutional Network (GCN) [25], Graph Attention Network (GAT) [46], and Graph Isomorphism Network (GIN) [55]. Interactions among distant atoms or functional groups within a molecule are crucial for effective molecular representation learning [53]. However, GNN-based methods fail to capture interactions among distant atoms or functional groups within molecules due to the over-squashing issue. This compression may result in the loss of crucial structural details, impairing the accuracy of molecular representations.

Inspired by KAN [24], which demonstrates robust data-fitting capabilities and high accuracy in small-scale AI + Science tasks, we believe that KAN is an ideal solution for addressing the challenge of insufficient labeled data in molecular representation learning. Furthermore, KAN's powerful data-fitting ability enables it to effectively capture crucial molecular structural details, mitigating the limitations of GNN-based architectures, such as the over-squashing problem. Several studies [6, 24, 61] have applied KAN to graph-based tasks, highlighting its potential. However, these efforts often

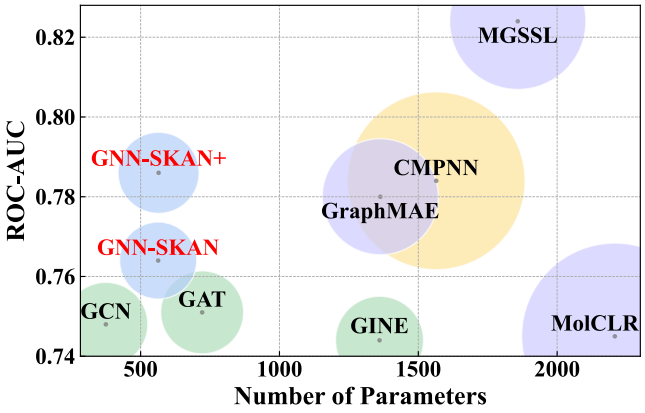


Figure 1: The visualization of our models' performance compared to the SOTAs on the HIV dataset. The circle size represents per-epoch computational cost. Purple circles represent self-supervised methods, green and yellow circles represent supervised methods, and blue circles represent our proposed methods (GNN-SKAN and GNN-SKAN+).

overlook the efficiency challenges associated with KAN, which limit its adaptation for molecular representation learning. As a result, integrating KAN into standard GNN architectures for molecular representation learning remains an open problem.

To bridge this gap, we introduce a new class of GNNs, **GNN-SKAN**, and its augmented variant, **GNN-SKAN+**, tailored for molecular representation learning. Our key innovation lies in adapting KAN to molecular graphs by developing a variant KAN that preserves KAN's high accuracy in small-scale tasks while improving efficiency and flexibility in handling diverse molecular structures. Herein, we propose **SwallowKAN** (**SKAN**), which replaces B-splines with adaptive RBFs to achieve these goals. By combining the strengths of SKAN and GNNs, we present a robust and versatile approach to molecular representation learning.

Specifically, we design a hybrid architecture that integrates GNN with SKAN, using a KAN-enhanced message-passing mechanism. To ensure versatility, SKAN is integrated into the update function to boost the expressiveness of the message-passing process. Additionally, we introduce GNN-SKAN+ which employs SKAN as the classifier to further reinforce model performance. Notably, our approach holds three key benefits: 1) Superior Performance: both GNN-SKAN and GNN-SKAN+ demonstrate superior prediction ability, robust generalization to unseen molecular scaffolds, and versatile transferability across different GNN architectures. 2) Efficiency: Our models achieve comparable or superior results to the SOTA self-supervised methods, while requiring less computational time and fewer parameters (Figure 1). 3) Few-shot Learning Ability: GNN-SKAN shows great potential in few-shot learning scenarios, achieving an average improvement of 6.01% across few-shot benchmarks.

The main contributions of this work are summarized as follows:

- To the best of our knowledge, we are the first to integrate KAN's exceptional approximation capabilities with GNNs' strengths in handling molecular graphs. This integration has led us to propose a new class of GNNs, termed GNN-SKAN, and its augmented variant, GNN-SKAN+, for molecular representation learning.
- We propose SwallowKAN (SKAN), which introduces learnable RBFs as base function. This modification not only effectively addresses the slow speed of the original KAN but also significantly enhances the model's adaptability to diverse data distributions.
- We conduct a theoretical analysis of the parameter counts and computational complexity of KAN and SKAN. Our comparative analysis shows that SKAN achieves higher computational efficiency.
- Our method achieves high accuracy and robust generalization on molecular property prediction tasks, surpassing or matching the SOTA models with lower time and memory requirements (Figure 1). It also excels in few-shot scenarios, with an average improvement of 6.01% across 4 few-shot learning benchmarks (Section 4).

2 Related Work

2.1 Molecular Representation Learning.

Existing molecular representation learning approaches can be broadly classified into three categories: 1) GNN-based approaches: Methods like MolCLR [49] and KANO [16] leverage contrastive learning framework, while MGSSL [62] and GraphMAE [20] employ generative pre-training strategies for molecular representation learning. Although these methods effectively learn molecular representations, they face the challenge of over-squashing [4]. 2) Transformer-based approaches: Models such as MolBERT [13], ChemBERTa [9], ChemBERTa-2 [1], SELFormer [60], and MolGen [15] enhance molecular representation learning with the self-attention mechanism, but they suffer from the quadratic complexity [45]. 3) Combination of both (Graph Transformers): These models, including Graphormer [58], GPS [36], Grover [37], and Uni-Mol [63], which combine message-passing neural networks with Transformer to enhance molecular representations. However, they also suffer from high computational complexity [23]. Unlike previous methods, we uniquely integrate GNNs with SKAN, a proposed KAN variant, to overcome the challenges present in existing molecular representation learning approaches.

2.2 Kolmogorov-Arnold Networks (KANs).

Recently, Liu et al. introduce the Kolmogorov-Arnold Network (KAN), which has been widely acknowledged for its high accuracy in small-scale AI + Science tasks. Unlike Multi-Layer Perceptrons (MLPs) [10], which use fixed activation functions on nodes ("neurons"), KAN employs learnable activation functions on edges ("weights"). This distinctive design significantly enhances KAN's performance, achieving up to 100 times greater accuracy than a four-layer MLP when solving partial differential equations (PDEs) with a two-layer structure [31]. Additionally, KAN has demonstrated exceptional performance across various tasks, including image recognition [3, 50], image generation [27], and time series prediction tasks [44, 54]. Recent work has extended KAN to the graph-structured domain. For instance, GKAN [12, 24], KAGNNs [6], and GraphKAN [61] integrate KAN into GNNs to enhance

performance. However, these approaches do not address KAN's efficiency issues. For example, GKAN incurs computational costs approximately 100 times higher than the standard GNNs on the Cora dataset [12], significantly hindering the broader adoption of KAN in graph-based applications. To improve the efficiency of KAN, several approaches have been proposed to replace B-splines with simpler base functions [42]. Notably, FastKAN [28] addresses this issue by replacing B-splines with RBFs [52], and GP-KAN [7] employs Gaussian Processes (GPs) [39] as an alternative, particularly for image classification tasks. Unlike these approaches, we introduce SKAN, a new KAN variant that incorporates adaptable RBFs specifically designed for efficient molecular representation learning. The differences between KAN, FastKAN and our SKAN are shown in Figure 2.

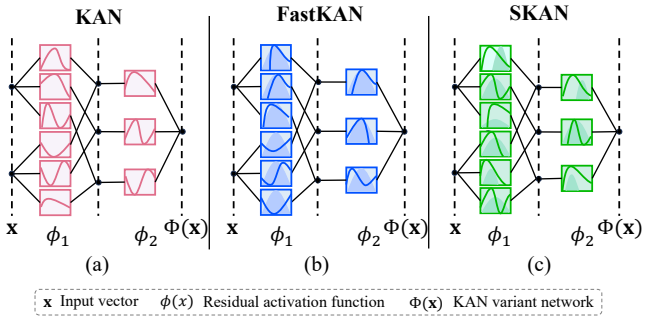


Figure 2: Comparison of three Kolmogorov-Arnold Network (KAN) variants, including (a) KAN [31] designed with B-splines (no shading), (b) FastKAN [28] designed with RBFs (fixed light blue shading), and (c) SwallowKAN (Our SKAN) designed with adaptable RBFs (adjustable light green shading).

3 Method

In this work, we generalize KAN to molecular graphs for molecular representation learning. The framework of our GNN-SKAN and GNN-SKAN+ are illustrated in Figure 3. This section focuses on the details of the implementation of our method. These details are crucial for ensuring the speed and reliability of the network.

3.1 Preliminary

Notations. Let $G = (\mathcal{V}, \mathcal{E})$ be a molecular graph, where $\mathcal{V}$ represents the set of nodes and $\mathcal{E}$ represents the set of edges. The number of nodes is $N = |\mathcal{V}|$. Let $\mathbf{H} = \{\mathbf{h}_v^{(0)} | v \in \mathcal{V}\}$ denote the initial features of nodes, and $\mathbf{E} = \{\mathbf{b}_{uv} | (u, v) \in \mathcal{E}\}$ denote the edge features. Let $\mathbf{z}$ represent the graph-level representation, $\hat{y}$ represent the graph-level prediction, and y represent the molecular property label (i.e., the ground-truth label). For the RBF, c_w and bw_w are the learnable center and bandwidth, respectively.

Kolmogorov-Arnold Representation Theorem (KART). KART [2, 5, 26] states that any multivariate continuous function defined on a bounded domain can be expressed as a finite composition of continuous single-variable functions and addition operations. Formally, for a smooth function $f : [0, 1]^n \to \mathbb{R}$ with n variables

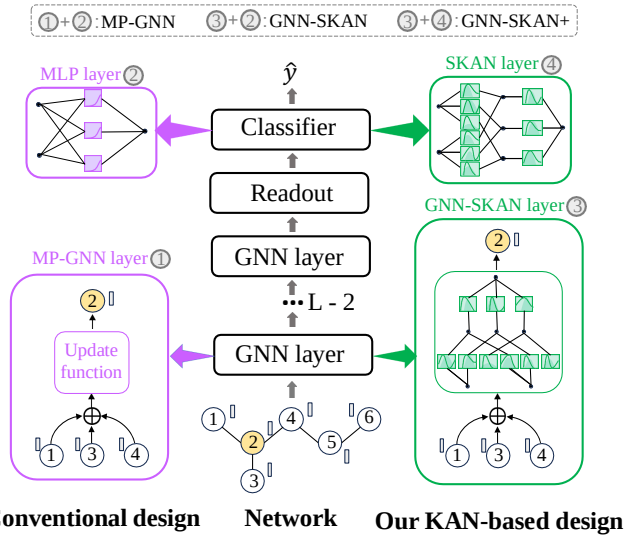


Figure 3: Comparison of the network architecture between MP-GNNs (left) and our SKAN-based designs (right), GNN-SKAN and GNN-SKAN+.

$\mathbf{x} = x_1, x_2, \ldots, x_n$, the theorem guarantees the existence of the following representation:

$$f(x_1, \ldots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^{n} \varphi_{q,p}(x_p) \right), \tag{1}$$

where $\Phi_q : \mathbb{R} \to \mathbb{R}$ and $\varphi_{q,p} : [0, 1] \to \mathbb{R}$.

The Design of KAN. Unlike the simple 2-layer KAN described by Eq. 1, which follows a $[n, 2n+1, 1]$ structure, KAN can be generalized to arbitrary widths and depths. Specifically, KAN consists of L layers and obtains outputs based on the input vector $\mathbf{x} \in \mathbb{R}^n$, as shown in the following equation:

$$\mathrm{KAN}(\mathbf{x}) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \cdots \circ \Phi_1 \circ \Phi_0)\, \mathbf{x}, \tag{2}$$

where Φ_l represents the function matrix of the lth layer. Each layer of KAN, defined by the function matrix Φ_l, is formed as follows:

$$\mathbf{x}_{l+1} = \underbrace{\begin{pmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,1,2}(\cdot) & \cdots & \phi_{l,1,n_l}(\cdot) \\ \phi_{l,2,1}(\cdot) & \phi_{l,2,2}(\cdot) & \cdots & \phi_{l,2,n_l}(\cdot) \\ \vdots & \vdots & & \vdots \\ \phi_{l,n_{l+1},1}(\cdot) & \phi_{l,n_{l+1},2}(\cdot) & \cdots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix}}_{\Phi_l} \mathbf{x}_l, \tag{3}$$

where $\phi_{l,j,i}$ denotes the activation function that connects neuron i in layer l to neuron j in layer $l + 1$. The network utilizes $n_l \times n_{l+1}$ activation functions between each pair of layers.

KAN enhances its learning capability by utilizing residual activation functions, $\phi(x)$, that combine a basis function and a spline function, as follows:

$$\phi(x) = w_b b(x) + w_s \,\mathrm{Spline}(x), \tag{4}$$

where $b(x)$ represents the $silu(x)$ function, and $\text{Spline}(x)$ denotes a linear combination of B-splines [43]. The parameters w_b and w_s are learnable.

3.2 Architecture

To address KAN's limitations in computational efficiency and adaptability, we introduce SwallowKAN (SKAN), a variant of KAN that employs learnable RBFs as activation functions. SKAN adaptively adjusts its parameters based on the data distribution, enabling effective modeling of molecules with diverse scaffolds. Building on the strengths of SKAN, we propose a new class of GNNs, GNN-SKAN, and its advanced version, GNN-SKAN+, which use a KAN-enhanced message-passing mechanism to enhance molecular representation learning.

GNN-based methods for molecular representation learning generally involves three key steps: aggregation, update, and readout. Specifically, SKAN is integrated into the update function, facilitating seamless incorporation into any MP-GNN. In GNN-SKAN+, SKAN also serves as the classifier to further improve performance. The complete algorithm of our approach is shown in Algorithm 1.

Step 1: Aggregation. In a molecular graph $G = (\mathcal{V}, \mathcal{E})$, let $\mathbf{h}^{(0)} \in \mathbb{R}^n$ represent the initial node features and $\mathbf{b} \in \mathbb{R}^n$ represent the initial edge features. At the dth layer, the GNN utilizes an aggregation function to combine information from the neighboring nodes of node v, forming a new message representation $\mathbf{m}_v^{(d)} \in \mathbb{R}^n$:

$$\mathbf{m}_v^{(d)} = \text{AGGREGATE}^{(d)}\left(\left\{\left(\mathbf{h}_v^{(d-1)}, \mathbf{h}_u^{(d-1)}, \mathbf{b}_{vu}\right) \mid u \in \mathcal{N}(v)\right\}\right), \tag{5}$$

where $u \in \mathcal{N}(v)$ denotes the set of neighboring nodes of node v.

Step 2: Update. After computing the aggregated message $\mathbf{m}_v^{(d)}$ for node v, the message is used to update the representation of node v at layer d:

$$\mathbf{h}_v^{(d)} = \text{UPDATE}^{(d)}\left(\mathbf{h}_v^{(d-1)}, \mathbf{m}_v^{(d)}\right). \tag{6}$$

To efficiently adapt to various data distributions, we design an augmented KAN, SwallowKAN (SKAN), which replace the original update function with learnable RBFs. The data center c_w and the bandwidth bw_w allow for efficient adaptation to the data distribution. The update function can be rewritten as follows:

$$\mathbf{h}_v^{(d)} = \text{SKAN}^{(d)}\left(\left(1 + \epsilon^{(d)}\right) \cdot \mathbf{h}_v^{(d-1)} + \mathbf{m}_v^{(d)}\right), \tag{7}$$

where $\epsilon^{(d)}$ is a parameter. Specifically, let $\mathbf{x} = \left(1 + \epsilon^{(d)}\right) \cdot \mathbf{h}_v^{(d-1)} + \mathbf{m}_v^{(d)}$. The residual activation function of SKAN at the dth layer is defined as:

$$\phi(x) = w_b b(x) + w_r \exp\left(-\frac{1}{2}\left(\frac{x - c_w}{bw_w}\right)^2\right)^1. \tag{8}$$

SKAN's adaptive parameter adjustment allows it to effectively handle the varied nature of molecular structures, leading to a more robust and adaptable model with improved generalization across diverse molecular datasets.

[1]For simplicity, each variable in this equation represent a single variable without a subscript.

Step 3: Readout. After performing Steps 1 and 2 for D iterations, we use the MEAN function as the readout mechanism to obtain the molecular representations. The readout operation is defined as:

$$\mathbf{g} = \text{READOUT}\left(\left\{\mathbf{h}_v^{(D)} \mid v \in \mathcal{V}\right\}\right), \tag{9}$$

where $\mathbf{g} \in \mathbb{R}^n$ represents molecular representation.

Step 4: Prediction. Next, we use an MLP to obtain the final prediction $\hat{y}$:

$$\hat{y} = \text{MLP}(\mathbf{g}) \in \mathbb{R}^{n_t}, \tag{10}$$

where n_t denotes the number of tasks in the molecular property classification or regression datasets. For the advanced version, GNN-SKAN+, we utilize a 2-layer SKAN as the classifier:

$$\hat{y} = \text{SKAN}(\mathbf{g}) \in \mathbb{R}^{n_t}. \tag{11}$$

The procedure of our method is summarized as follows:

Algorithm 1 GNN-SKAN and GNN-SKAN+ for Molecular representation Learning

Input: Molecular graph $G = (\mathcal{V}, \mathcal{E})$ with initial node features $\mathbf{H} = \{\mathbf{h}_v^{(0)} | v \in \mathcal{V}\}$ and edge features $\mathbf{E} = \{\mathbf{b}_{uv} | (u, v) \in \mathcal{E}\}$, number of layers D
Output: Molecular property prediction $\hat{y}$
Initialize node representations: $\mathbf{h}_v^{(0)}$ for all $v \in V$
for each iteration $d = 1$ to D **do**
  for each node $v \in V$ **do**
    Aggregate messages from neighbors by Eq. 5
  end for
  for each node $v \in V$ **do**
    Update node representation by Eq. 7
  end for
end for
Perform readout for molecular representation by Eq. 9
Predict molecular property by Eq. 10 (for GNN-SKAN) or Eq. 11 (for GNN-SKAN+)
return $\hat{y}$

3.3 The Efficiency of SKAN

For a SKAN network, we denote L as the number of layers in the network, N as the number of nodes in each layer, and M as the number of RBFs.

Parameter Complexity. In the original KAN, the total number of parameters is $O(LN^2(G + k))$, where G represents the number of grid points and k (typically $k = 3$) is the order of each spline. Since k is much smaller than G [31] and its impact on the total number of parameters is usually negligible, $O(N^2L(G + k))$ simplifies to $O(N^2LG)$. In comparison, SKAN has a total of $O(LN^2M)$ parameters, where $M \leq 8$, which is smaller than G. As a result, SKAN requires fewer parameters than the original KAN, improving model efficiency while maintaining expressive capacity.

Table 1: Comparison with the base MP-GNNs on six molecular classification benchmarks. The best results are in bold. The mean and standard deviation of ROC-AUC over five independent test runs are reported. '-' indicates out of memory.

Model	BBBP ↑	Tox21 ↑	ToxCast ↑	SIDER ↑	HIV ↑	BACE ↑
GCN	0.622 ± 0.037	0.738 ± 0.003	0.624 ± 0.006	0.611 ± 0.012	0.748 ± 0.007	0.696 ± 0.048
GCN-GKAN	-	0.742 ± 0.005	-	-	-	0.732 ± 0.013
GCN-SKAN	0.652 ± 0.009	0.737 ± 0.003	0.632 ± 0.006	0.603 ± 0.002	0.757 ± 0.021	**0.754 ± 0.023**
GCN-SKAN+	**0.676 ± 0.014**	**0.747 ± 0.005**	**0.643 ± 0.004**	**0.614 ± 0.005**	**0.786 ± 0.015**	0.747 ± 0.009
GAT	0.640 ± 0.032	0.732 ± 0.003	0.635 ± 0.012	0.578 ± 0.017	0.751 ± 0.010	0.653 ± 0.016
GAT-GKAN	-	0.725 ± 0.008	-	-	-	0.712 ± 0.014
GAT-SKAN	0.649 ± 0.007	0.723 ± 0.005	0.633 ± 0.003	0.603 ± 0.004	0.746 ± 0.016	**0.755 ± 0.028**
GAT-SKAN+	**0.688 ± 0.015**	**0.748 ± 0.004**	**0.640 ± 0.004**	**0.620 ± 0.002**	**0.784 ± 0.004**	0.731 ± 0.060
GINE	0.644 ± 0.018	0.733 ± 0.009	0.612 ± 0.009	0.596 ± 0.003	0.744 ± 0.014	0.612 ± 0.056
GINE-GKAN	-	0.734 ± 0.011	-	-	-	0.706 ± 0.018
GINE-SKAN	0.652 ± 0.012	0.738 ± 0.008	0.627 ± 0.005	0.586 ± 0.004	0.764 ± 0.008	**0.768 ± 0.011**
GINE-SKAN+	**0.688 ± 0.016**	**0.750 ± 0.002**	**0.630 ± 0.005**	**0.613 ± 0.004**	**0.785 ± 0.009**	0.762 ± 0.027

Computational Complexity. The computational complexity of the original KAN is $O((LN^2)k)$, where k denotes the order of the spline. In contrast, SKAN achieves a lower computational complexity of $O(LN^2)$ since each RBF operates with a complexity of $O(1)$.

Overall, SKAN demonstrates significant advantages over the original KAN in terms of parameter efficiency and computational scalability. Notably, despite utilizing two layers, our GNN-SKAN and GNN-SKAN+ outperform 5-layer baseline GNNs and achieve performance comparable to or exceeding state-of-the-art self-supervised methods, while maintaining higher efficiency.

4 Experiments

Datasets. We evaluate our GNN-SKAN and GNN-SKAN+ across a wide range of public molecular property prediction benchmarks. The evaluation includes three categories: 1) **Molecular Classification Datasets**: BBBP, Tox21, ToxCast, SIDER, HIV, and BACE; 2) **Molecular Regression Datasets**: Lipo, FreeSolv, Esol, QM7, QM8, and QM9; and 3) **Few-shot Learning Datasets**: Tox21, SIDER, MUV, and ToxCast from PAR [47]. Further dataset details are provided in Appendix A.

Experimental Setup. Following the experimental protocols established by Hu* et al. [21] and Hou et al. [20], we use ROC-AUC as the evaluation metric for classification tasks and mean absolute error (MAE) for regression tasks. The results are reported as the mean and standard deviation across five independent test runs. Our model employs a 2-layer GNN-SKAN encoder with 256 hidden units, paired with an MLP classifier. All baseline models use a 5-layer GNN architecture. Additionally, we introduce GNN-SKAN+, which incorporates a 2-layer SKAN classifier. The learning rate ranges from 0.01 to 0.001, and the number of adaptive RBFs ranges from 3 to 8. All experiments are conducted on a single NVIDIA RTX A6000 GPU.

4.1 Classification Tasks

Comparison with Baseline MP-GNNs. Table 1 summarizes the performance of three baseline MP-GNNs (GCN [25], GAT [46], and GINE [21]) with their augmented versions, including GKAN [24], SKAN, and SKAN+, across six classification benchmarks. Given the architectural similarities among GKAN, GraphKAN [61], and KAGNNs [6], we extend the recently proposed GKAN as a baseline to other MP-GNNs, which integrate GNNs with KAN. In Table 1, our augmented models, GNN-SKAN and GNN-SKAN+, consistently outperform the base MP-GNNs and GNN-GKAN across most benchmarks. Notably, GCN-SKAN improves by 4.82% over GCN, and GCN-SKAN+ improves by 8.68% over GCN on the BBBP dataset. Using scaffold splitting [33, 56] to evaluate generalization, the augmented models show outstanding performance on new scaffolds. For instance, GINE-SKAN outperforms GINE by 25.49% on the BACE dataset, demonstrating robust generalization. These results indicate that the augmented models effectively capture structural information of molecules and adapt to diverse molecular scaffolds. Additionally, our augmented models demonstrate outstanding performance across different baseline MP-GNNs, highlighting their transferability and versatility.

Comparison with SOTAs. Table 2 presents a comparison of the best-performing augmented models from Table 1 with state-of-the-art methods across six molecular classification datasets. The baselines are divided into three categories: 1) supervised methods: GCN [25], GAT [46], GINE [21], and CMPNN [41]; 2) Self-supervised methods: MolCLR [49], MGSSL [62], and GraphMAE [20]; 3) Graph Transformer-based methods: Grover$_{base}$ [37] and GPS [36]. The results demonstrate that our augmented models match or surpass the SOTA performance. In particular, GNN-SKAN and GNN-SKAN+ significantly outperform supervised methods such as GCN, GAT, and CMPNN, especially on datasets like Tox21 and BACE, highlighting the enhanced ability of our models to capture molecular representations. Additionally, when compared to self-supervised methods such as MolCLR, MGSSL, and GraphMAE, GNN-SKAN and GNN-SKAN+ again demonstrate competitive performance, showcasing the robustness and flexibility of our approach in various settings. Notably, GNN-SKAN+ achieves the highest test ROC-AUC

Table 2: Comparison of our best results from Table 1 with the SOTA models on six classification benchmarks of physiology and biophysics. The best results are indicated in bold. The mean and standard deviation of test ROC-AUC on five independent runs are reported.

Model	BBBP ↑	Tox21 ↑	ToxCast ↑	SIDER ↑	HIV ↑	BACE ↑
GCN	0.622 ± 0.037	0.738 ± 0.003	0.624 ± 0.006	0.611 ± 0.012	0.748 ± 0.007	0.696 ± 0.048
GAT	0.640 ± 0.032	0.732 ± 0.003	0.635 ± 0.012	0.578 ± 0.017	0.751 ± 0.010	0.653 ± 0.016
GINE	0.644 ± 0.018	0.733 ± 0.009	0.612 ± 0.009	0.596 ± 0.003	0.744 ± 0.014	0.612 ± 0.056
CMPNN	**0.737 ± 0.065**	0.709 ± 0.024	**0.708 ± 0.013**	0.548 ± 0.020	0.782 ± 0.022	0.784 ± 0.026
MolCLR	0.706 ± 0.031	0.747 ± 0.080	0.659 ± 0.021	0.612 ± 0.036	0.745 ± 0.210	0.823 ± 0.012
MGSSL	0.724 ± 0.070	0.745 ± 0.026	0.638 ± 0.023	0.578 ± 0.036	0.774 ± 0.006	0.824 ± 0.021
GraphMAE	0.706 ± 0.060	0.744 ± 0.034	0.641 ± 0.003	0.603 ± 0.011	0.772 ± 0.010	0.780 ± 0.012
Grover_{base}	0.700 ± 0.009	0.743 ± 0.012	0.654 ± 0.006	**0.656 ± 0.005**	0.776 ± 0.023	**0.826 ± 0.016**
GPS	0.698 ± 0.012	0.748 ± 0.006	0.655 ± 0.014	0.645 ± 0.008	0.774 ± 0.005	0.819 ± 0.011
GNN-SKAN	0.652 ± 0.009	0.738 ± 0.008	0.633 ± 0.003	0.603 ± 0.004	0.764 ± 0.008	0.768 ± 0.011
GNN-SKAN+	0.688 ± 0.015	**0.750 ± 0.002**	0.643 ± 0.004	0.620 ± 0.002	**0.786 ± 0.015**	0.762 ± 0.027

Table 3: Comparison with standard GNNs on six molecular regression benchmarks. The best results are indicated in bold. The mean and standard deviation of the test mean absolute error on five independent runs are reported.

Model	Lipo ↓	FreeSolv ↓	Esol ↓	QM7 ↓	QM8 ↓	QM9 ↓
GCN	0.809 ± 0.053	2.891 ± 0.249	**1.241 ± 0.086**	79.393 ± 2.789	0.025 ± 0.001	0.311 ± 0.008
GCN-SKAN	0.921 ± 0.064	**1.888 ± 0.043**	1.261 ± 0.049	79.443 ± 0.649	**0.023 ± 0.000**	0.266 ± 0.004
GCN-SKAN+	**0.771 ± 0.015**	1.893 ± 0.052	1.410 ± 0.074	**77.512 ± 0.987**	**0.023 ± 0.000**	**0.260 ± 0.007**
GAT	0.800 ± 0.025	2.360 ± 0.501	**1.345 ± 0.095**	112.989 ± 4.884	0.025 ± 0.001	0.380 ± 0.008
GAT-SKAN	0.824 ± 0.015	**1.935 ± 0.123**	1.348 ± 0.089	105.494 ± 4.758	0.026 ± 0.001	0.374 ± 0.007
GAT-SKAN+	**0.777 ± 0.020**	2.077 ± 0.174	1.347 ± 0.079	**100.664 ± 3.939**	**0.024 ± 0.000**	**0.371 ± 0.010**
GINE	1.158 ± 0.227	2.086 ± 0.092	1.275 ± 0.077	94.559 ± 23.708	0.034 ± 0.013	0.347 ± 0.013
GINE-SKAN	0.934 ± 0.121	1.722 ± 0.125	**1.098 ± 0.018**	78.200 ± 1.536	**0.022 ± 0.000**	**0.250 ± 0.004**
GINE-SKAN+	**0.782 ± 0.002**	**1.715 ± 0.130**	1.175 ± 0.022	**77.315 ± 2.114**	0.024 ± 0.001	0.254 ± 0.002

Table 4: Comparison of common few-shot learning methods on four few-shot molecular property prediction benchmarks. The best results are in bold. Mean and standard deviation of test ROC-AUC across five independent runs.

Model	#Param.	Tox21 (10-shot)	SIDER (10-shot)	MUV (10-shot)	ToxCast (10-shot)
TPN	2,076.73K	0.795 ± 0.002	0.675 ± 0.003	0.783 ± 0.007	0.708 ± 0.006
TPN-SKAN	982.51K	**0.834 ± 0.002**	**0.799 ± 0.004**	**0.835 ± 0.005**	**0.755 ± 0.001**
MAML	1,814.94K	0.762 ± 0.001	0.707 ± 0.001	**0.814 ± 0.002**	0.699 ± 0.017
MAML-SKAN	736.55K	**0.809 ± 0.002**	**0.828 ± 0.001**	0.784 ± 0.002	**0.703 ± 0.015**
ProtoNet	1,814.36K	0.634 ± 0.001	**0.592 ± 0.001**	**0.719 ± 0.001**	0.637 ± 0.015
ProtoNet-SKAN	732.63K	**0.705 ± 0.002**	0.589 ± 0.001	0.705 ± 0.002	**0.639 ± 0.012**
PAR	2,377.91K	0.804 ± 0.002	0.680 ± 0.001	0.780 ± 0.001	0.698 ± 0.004
PAR-SKAN	1,124.70K	**0.830 ± 0.003**	**0.829 ± 0.001**	**0.831 ± 0.001**	**0.751 ± 0.002**

on the HIV dataset, outperforming GraphMAE by a significant margin. Similarly, GNN-SKAN shows competitive results on the HIV and BACE datasets, surpassing several other methods.

Figure 1 presents a comparison of the best-performing augmented models from Table 1 with the SOTAs on the HIV dataset. In Figure 1, our models, GNN-SKAN and GNN-SKAN+, consistently outperform or match the state-of-the-art methods while demonstrating superior efficiency. Notably, our models achieve a substantial reduction in parameters count—approximately 563K and 565K, respectively—compared to GraphMAE (∼1363K) and MGSSL (∼1858K), while maintaining comparable performance. Moreover,

GNN-SKAN and GNN-SKAN+ show a faster training speeds per epoch. In contrast, models with a higher parameter count, such as GraphMAE and MolCLR, require significantly longer training times. This highlights the efficiency of our models, making them more suitable for practical applications. Additional visualizations on other datasets are provided in Appendix B.

4.2 Regression Tasks

Table 3 presents a comparison between the baseline MP-GNNs and their augmented counterparts, GNN-SKAN and GNN-SKAN+, across six regression benchmarks in physical chemistry and quantum mechanics. The augmented models consistently outperform their base counterparts across most datasets, highlighting the effectiveness of SKAN. Remarkable advancements are observed on the Lipo and FreeSolv datasets. Specifically, GCN-SKAN+ and GINE-SKAN+ show improvements of 4.70% and 32.47% on the Lipo dataset, and 34.52% and 17.78% on the FreeSolv dataset, respectively. In summary, these results underscore the potential of GNN-SKAN and GNN-SKAN+ to significantly enhance the accuracy and reliability of GNNs for molecular regression tasks.

4.3 Few-shot Learning Tasks

Table 4 demonstrates the effectiveness of SKAN in addressing the few-shot learning problem. We select four common few-shot baselines: TPN [30], MAML [17], ProtoNet [40], and PAR [48]. All experiments are conducted in a 2-way 10-shot setting. We do not include 1-shot experiments, as they are less meaningful in drug discovery. As shown in Table 4, the augmented models consistently outperform their base counterparts across most datasets, with average improvements of 9.14% for TPN-SKAN, 5.04% for MAML-SKAN, 2.26% for ProtoNet-SKAN, and 7.59% for PAR-SKAN. These enhancements, achieved through the integration of the SKAN architecture, result in higher ROC-AUC values, significantly improving the accuracy and reliability of few-shot molecular property prediction methods.

4.4 Visualization

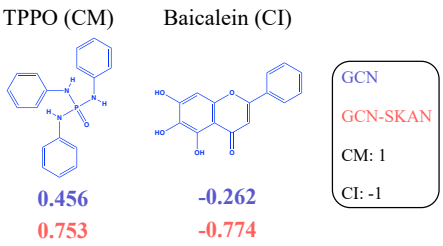


Figure 4: Predicted scores for TPPO (CM) and Baicalein (CI) using GCN and GCN-SKAN. The CM class is labeled as 1, and the CI class as -1.

The examples in Figure 4 illustrate that GCN-SKAN exhibits higher confidence in its predictions for both TPPO and Baicalein on the HIV dataset. This further indicates that SKAN improves the model's generalization ability across diverse molecular structures, thereby improving overall predictive performance.

Table 5: Ablation study of SKAN in Aggregation and Update functions and classifier with GINE on the HIV dataset. The highlights are the first, second, and third best results.

Agg. with SKAN	Update with SKAN	Classifier with SKAN	Result ROC-AUC
✗	✗	✗	0.744 ± 0.014
✓	✗	✗	0.746 ± 0.014
✗	✓	✗	0.764 ± 0.010[a]
✗	✗	✓	0.762 ± 0.008
✓	✓	✗	0.767 ± 0.009
✓	✗	✓	0.761 ± 0.015
✗	✓	✓	0.785 ± 0.009
✓	✓	✓	0.782 ± 0.011[b]

[a] GINE-SKAN. [b] GINE-SKAN+.

4.5 Ablation Study

We conduct ablation studies to evaluate the design choices for each component of proposed architecture. Specifically, we evaluate the effectiveness of SKAN and its integration within GNNs. Additionally, we explore the impact of different SKAN configurations, including the number of SAKN-GNN layers and the number of SKAN layers. Finally, the hyperparameter selections for our method are provided in Appendix C.

The Effectiveness of SKAN. Figure 5 (a) compares the performance of GINE and its KAN variants across three datasets. GINE-SKAN shows notably superior performance over GINE and other variants, such as GINE-KAN and GINE-FastKAN. The adaptive bandwidths and centers in SKAN enhance its ability to adapt to data distribution, thereby improving flexibility. Figure 5 (b) presents the computational efficiency analysis. While GINE-KAN incurs significantly higher computation time per epoch compared to the base model (GINE), our model exhibits lower computation time, highlighting its efficiency.

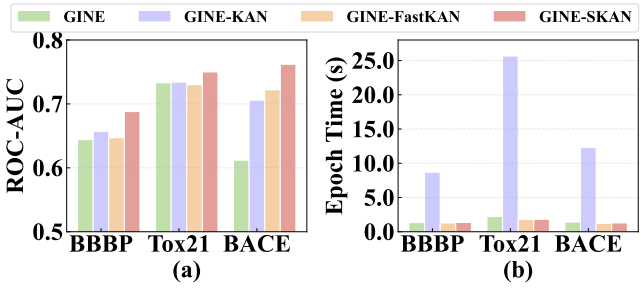


Figure 5: The comparison of (a) performance and (b) time cost between GINE and its augmented versions, including Original KAN, FastKAN, SKAN, across three datasets (BBBP, Tox21, BACE).

SKAN Integration in GNNs. Table 5 compares the performance of GINE with SKAN integrated into different components (Aggregation and Update, and Classifier) on the HIV dataset. The results are presented as the mean and standard deviation of the test ROC-AUC across five runs. The results indicate that integrating SKAN into the update function (GINE-SKAN) is more effective than integrating it into the aggregation function (GINE-SKAN (Agg.)). This advantage is likely due to the update function's role in executing nonlinear feature transformations at each GNN layer. Additionally, using SKAN as the classifier, our model (GINE-SKAN+) achieves SOTA performance.

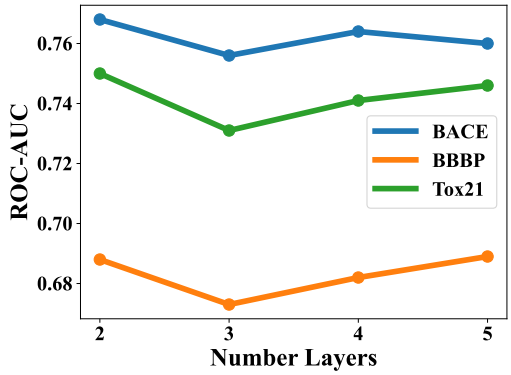


Figure 6: The performance of increasing number of GINE-SKAN layers.

Architectural Layers. Figure 6 illustrates the impact of increasing the number of GINE-SKAN layers on model performance. On the BACE dataset, our model consistently showing higher ROC-AUC values across all layer configurations. On the Tox21 dataset, our model exhibits stable performance, with a slight dip at 3 layers. Meanwhile, on the BBBP dataset, our model shows an increase in performance as the number of layers increases. Based on these findings, we select 2 layers as the optimal setting for GINE-SKAN.

Figure 7 demonstrates that increasing the number of SKAN layers in the GINE-SKAN model does not necessarily lead to better performance. For all three datasets, there is a general trend of declining ROC-AUC scores as the model becomes deeper. This suggests that while a modest number of layers (around 2-3) may be beneficial, further increasing the depth of the model could introduce overfitting or unnecessary complexity, ultimately reducing the model's effectiveness. Therefore, careful consideration of the number of layers is crucial when designing GINE-SKAN architectures for different datasets. Based on these observations, we opt for 2 layers as the optimal configuration.

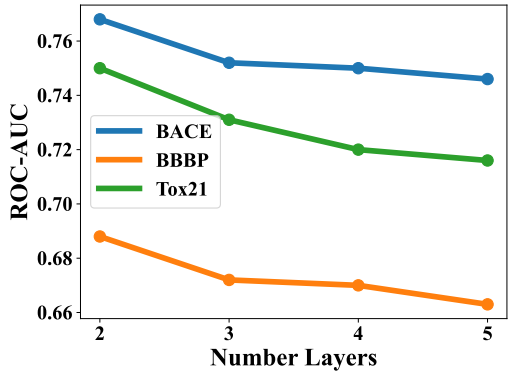


Figure 7: The performance of increasing SKAN layers.

5 Conclusion

This work introduces GNN-SKAN and GNN-SKAN+, a new class of GNNs that integrate traditional GNN architectures with KAN. The core of our approach is SwallowKAN (SKAN), a novel KAN variant specifically designed to address the challenge of molecular diversity and enhance computational efficiency. We evaluate our models on widely used molecular benchmark datasets, including six classification, six regression, and four few-shot learning datasets. The results show that our approach significantly enhances molecular representation learning. For future work, we aim to explore the integration of Graph Transformer architectures with KAN to address the high computational complexity inherent in these models, potentially further advancing molecular representation learning.

Limitation. This work focuses primarily on widely used MP-GNNs, without assessing the effectiveness of our approach on more advanced GNN architectures.. Additionally, we observe that SKAN-GNN demonstrates greater performance improvements in classification tasks compared to regression tasks. In future work, we plan to investigate the underlying reasons for this disparity and explore models better suited for regression tasks.

Acknowledgements

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Table 1: Detailed description of molecular classification and regression datasets.

Dataset	Type of the task	Content	# of compounds	Metric
BBBP	Binary classification	Binary labels on blood-brain barrier permeability	2,039	ROC-AUC
Tox21	Binary classification	Qualitative toxicity measurements on 12 targets	7,831	ROC-AUC
ToxCast	Binary classification	Qualitative toxicity measurements on 617 targets	8,597	ROC-AUC
SIDER	Binary classification	Classification of drug side-effects into 27 system organ classes	1,427	ROC-AUC
HIV	Binary classification	Binary labels on the ability to inhibit HIV replication	41,127	ROC-AUC
BACE	Binary classification	Binary labels on human-secretase 1 (BACE1) binding properties	1,513	ROC-AUC
Lipo	Regression	Experimental octanol/water distribution coefficient of compounds	4,200	MAE
FreeSolv	Regression	Hydration free energy of small molecules in water	642	MAE
Esol	Regression	Aqueous solubility of common small molecules	1,128	MAE
QM7	Regression	Prediction of atomization energies	7,165	MAE
QM8	Regression	Prediction of 12 quantum mechanical properties	21,786	MAE
QM9	Regression	Prediction of 12 quantum mechanical properties	134,000	MAE

Table 2: Summary of few-shot datasets used.

Dataset	# Compounds	# Tasks	# Meta-Training Tasks	# Meta-Testing Tasks
Tox21	8,014	12	9	3
SIDER	1,427	27	21	6
MUV	93,127	17	12	5
ToxCast	8,597	617	450	167

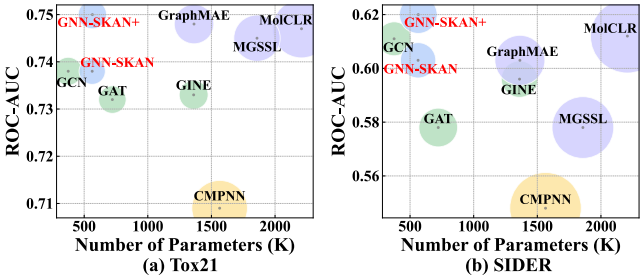


Figure 1: Performance comparison of our models with the SOTAs on the Tox21 and SIDER datasets. The x-axis represents the number of parameters (K), and the y-axis represents the ROC-AUC scores. Each circle's size corresponds to the per-epoch time. GNN-SKAN+ and GNN-SKAN are highlighted in red. Purple circles represent SSL methods, green and yellow circles represent supervised learning methods, and blue circles represent our methods.

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A Benchmark Datasets

Table 1 provides a detailed breakdown of various molecular datasets used for molecular classification and regression tasks. It lists the dataset name, the type of task (binary classification or regression), the content (a brief description of the specific task or property being predicted), the number of compounds included in each dataset, and the evaluation metric used (ROC-AUC for classification tasks and MAE for regression tasks).

Table 2 summarizes datasets that are employed in few-shot learning experiments [47]. It details the number of compounds, the total number of tasks, and further breaks down the tasks into meta-training and meta-testing tasks for each dataset. The few-shot datasets include Tox21, SIDER, MUV, and ToxCast, with varying numbers of compounds and tasks.

B Comparison with SOTAs

Figure 1 illustrates that GNN-SKAN+ is a strong contender in molecular classification tasks, offering a compelling balance of high performance, low parameter count, and computational efficiency. This makes it a versatile choice for applications where both accuracy and resource efficiency are critical. In contrast, while models like MolCLR [49] and GraphMAE [20] achieve high ROC-AUC scores, their increased complexity and computational costs may limit their practicality in some scenarios.

C Hyperparameter Settings

The hyperparameters selected for the GNN-SKAN model reflect a deliberate balance between simplicity and performance. As summarized in Table 3, the model uses 2 layers each for GNN-SKAN and SKAN, no dropout, and a hidden dimension of 256. This configuration is designed to avoid overfitting while still being capable of

Table 3: Hyperparameters used by GNN-SKAN.

Hyperparameter	Selected
learning rate	[0.001, 0.0001]
number of GNN-SKAN layers	2
number of SKAN layers	2
number of Adaptive RBFs	[3,4,5,6,7,8]
bandwidth	[2, 4, 6, 8]
dropout	0
hidden dimension for GNN-SKAN	256
optimizer	Adam

capturing complex patterns. The use of a low learning rate and the Adam optimizer further supports the goal of achieving stable and effective training. This setup suggests that the GNN-SKAN model is optimized for scenarios where model simplicity and computational efficiency are important, without sacrificing too much in terms of performance.

D Motivation

Molecular graphs exhibit a hierarchical structure, typically consisting of three levels: atoms, substructures (e.g., functional groups), and the entire molecule. Each level contributes distinctively to the molecular properties. In traditional Graph Neural Networks (GNNs), node representations are built hierarchically via local aggregation, but they struggle to adaptively capture the multi-scale semantics needed for fine-grained molecular property prediction.

To overcome these limitations, we propose the integration of SwallowKAN (SKAN) into the GNN update mechanism. SKAN utilizes adaptive Radial Basis Functions (RBFs) to enhance flexibility and expressivity in node updates, allowing the model to capture contextual molecular substructures more effectively.

We model a molecular graph as $G = (V, E)$, where each node $v \in V$ has a representation $\mathbf{h}_v^{(d)}$ at layer d. The node update in GNN-SKAN is defined as:

$$\mathbf{h}_v^{(d)} = \text{SKAN}^{(d)}\left((1+\epsilon^{(d)}) \cdot \mathbf{h}_v^{(d-1)} + \mathbf{m}_v^{(d)}\right)$$

where $\epsilon^{(d)}$ is a learnable residual coefficient, and $\mathbf{m}_v^{(d)}$ denotes the message aggregated from neighboring nodes.

The core of SKAN lies in its RBF-based activation function:

$$\phi(x) = w_b \cdot b(x) + w_r \cdot \exp\left(-\frac{1}{2}\left(\frac{x - c_w}{b_{ww}}\right)^2\right)$$

where c_w and b_{ww} are learnable center and bandwidth parameters. These enable SKAN to adaptively emphasize different substructure patterns, such as benzene rings or ester groups, by modifying the shape and position of the RBF kernel.

This hierarchical and adaptive update scheme makes GNN-SKAN particularly effective for molecular representation learning, enabling it to capture both local atomic features and global functional semantics.

E More Advanced GNN Experiments

To further evaluate the generalizability and adaptability of SKAN across diverse GNN backbones, we integrate the SKAN module into two widely used advanced architectures: AttentiveFP and 4-HiGCN. Specifically, we construct two SKAN-enhanced variants for each backbone: SKAN and SKAN+.

Experimental Setup. All models are trained and evaluated on six benchmark molecular classification datasets: BBBP, Tox21, ToxCast, SIDER, HIV, and BACE. We follow the same data splits and preprocessing pipelines as prior works. Each experiment is repeated with five different random seeds, and we report the mean and standard deviation of ROC-AUC scores.

Results and Observations. Table 4 shows the performance comparisons:

- Across all datasets, integrating SKAN improves performance over the original AttentiveFP and 4-HiGCN backbones.
- The **SKAN**$^+$ variant achieves the highest ROC-AUC in all six datasets for both architectures, indicating that the combined use of SKAN in both message and feature update layers significantly enhances model expressivity.
- Notably, on the ToxCast dataset, SKAN$^+$ improves AttentiveFP from 0.542 to 0.621, a relative gain of over 14.6%. Similarly, on the BACE dataset, SKAN$^+$ boosts performance from 0.812 to 0.845.
- These consistent gains demonstrate that SKAN provides a general, architecture-agnostic mechanism for improving molecular graph learning by enabling adaptive, structure-aware transformations.

Overall, these results validate the effectiveness of SKAN not only as a lightweight replacement for MLP layers but also as a universal plugin module for diverse GNN variants.

Table 4: Performance comparison of advanced GNNs with and without SKAN integration across multiple molecular classification tasks. Bold values indicate the best performance per dataset.

Model	BBBP	Tox21	ToxCast	SIDER	HIV	BACE
AttentiveFP	0.679 ± 0.021	0.732 ± 0.014	0.542 ± 0.025	0.605 ± 0.013	0.721 ± 0.023	0.812 ± 0.015
AttentiveFP-SKAN	0.684 ± 0.015	0.729 ± 0.034	0.549 ± 0.009	0.608 ± 0.018	0.715 ± 0.017	0.821 ± 0.027
AttentiveFP-SKAN$^+$	**0.706 ± 0.014**	**0.741 ± 0.024**	**0.621 ± 0.011**	**0.620 ± 0.021**	**0.734 ± 0.021**	**0.845 ± 0.012**
4-HiGCN	0.652 ± 0.016	0.741 ± 0.027	0.635 ± 0.014	0.628 ± 0.023	0.763 ± 0.016	0.637 ± 0.031
4-HiGCN-SKAN	0.671 ± 0.007	0.756 ± 0.015	0.645 ± 0.019	0.635 ± 0.031	0.754 ± 0.035	0.620 ± 0.013
4-HiGCN-SKAN$^+$	**0.688 ± 0.015**	**0.765 ± 0.021**	**0.663 ± 0.013**	**0.654 ± 0.011**	**0.769 ± 0.022**	**0.656 ± 0.027**